

LESSON 7.9

PERCENT INCREASE AND DECREASE



LESSON INTRODUCTION

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step, real-world percent problems.

LEARNING TARGETS

- I can find percent increase or percent decrease.

BENCHMARK COHERENCE

BUILDING ON

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

WORKING TOWARD

N/A

ESSENTIAL QUESTIONS

- How can you model percent change problems with a tape diagram and double number lines? How do you calculate percent increase and percent decrease?

LESSON NARRATIVE

Students solve a variety of real-world percent change problems. Initially, students use tape diagrams and double number lines to model the problems. Then, students are introduced to the percent of change formula and apply that to additional problems in the lesson.

COMPONENT SUMMARY

7.9.1 WARM-UP (~5 MIN.)

Use a percentage to determine the fewest number of people surveyed

7.9.3 TRY IT (10 MIN.)

Use models to represent percent change

7.9.5 YOUR TURN! (5 MIN.)

Additional practice solving problems involving percent change

7.9.2 GUIDED INSTRUCTION (10 MIN.)

Learn how to find the percent of change using models

7.9.4 COLLABORATION (10-15 MIN.)

Solve problems involving percent change

7.9.6 WRAP-UP (~5 MIN.)

Error analysis of percent change

RESOURCES

GLOSSARY

Key Terms:

- Percent Of change

Additional Terms:

- N/A

MATERIALS

- N/A

ADDITIONAL PREPARATION

- Consider creating and displaying an anchor chart with instructions, examples, and visuals for students to reference.

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle to understand why they can add the percent increase to 100% or subtract the percent decrease from 100%.
- Students may reverse the order of the values in the percent of change formula.

THINGS TO CONSIDER

- If students struggle with percent change formulas, it may help to walk them through the process to get to $\frac{\text{difference}}{\text{original}}$. This includes students recognizing that the percent of change will be positive for an increase and negative for a decrease.
- Consider allowing students to use calculators for the 7.9.4 Collaboration, 7.9.5 Your Turn!, and 7.9.6 Wrap-Up. Allowing students to use calculators during the lesson will help students focus on discovery and the process of calculating percent of change. Consider having no-calculator questions where students turn their calculator over. Teachers should look for and consider opportunities for students to improve their fluency with operations with rational numbers. For example, students should not need a calculator in the components prior to 7.9.4 Collaboration.

LITERACY AND LANGUAGE ROUTINES

Take Turns

For the 7.9.4 Collaboration, have students take turns explaining how they solved questions 1 and 2. Partners can provide feedback and compare their work. These discussions will help students gain confidence and understanding.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.4.1 Discussion and Reflection

The 7.9.6 Wrap-Up provides the opportunity for students to analyze the percent of change error in a sample of student work. Through identifying and correcting mistakes in a sample of student work, students will be less likely to make mistakes themselves.

DIFFERENTIATE INSTRUCTION

In 7.9.4 Collaboration, students transition into using the percent of change formula. If students are not ready for this abstraction, allow them to continue working with the pictorial representations. Help students see the connections between the models and the formula. Consider modeling how to use the formula, if needed, and discussing the connections between the formula and the models explicitly.

7.9.1 WARM-UP: INTERPRETING PERCENTAGES

TEACHER MOVES

This component is scaffolded such that parts could be released to students at this point in the unit. Consider which questions may need more modeling or elaboration and which would be best for students to complete with a partner.



7.9.1 Warm-Up: Interpreting Percentages

1. If exactly 93.6% of people completed a survey, what is the fewest number of people that were surveyed? Consider: How could it be determined if only one person could have completed it? Can partial people complete the survey?

125 is the fewest number of people that could have been surveyed. If one person completed the survey, then the percentage would have to be 100%. You cannot have a fractional part of a person complete a survey. For exactly 93.6% of people to complete a survey, the amount of people surveyed must be at least 125.

2. Explain your answer using mathematics.

I rewrote the percentage as a fraction and then used equivalent fractions to eliminate the decimal by putting the percentage over 1000 since I can not have a decimal amount of a person. Then, I reduced it to lowest terms.

$$\frac{93.6}{100} \times \frac{10}{10} = \frac{936}{1000} \div \frac{8}{8} = \frac{117}{125}$$



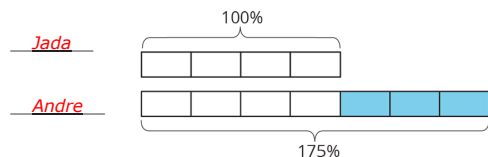
7.9.2 Guided Instruction: Percent of Change

Percentages can be used to make comparisons from one quantity to another.

- It takes Andre $\frac{3}{4}$ times longer than it took Jada to get to school.

The diagram models this in terms of percentages.

- Label each line of the tape diagram with the name of the student who is represented.



- Complete the statements.

The time it takes Andre to get to school is 75% ☐ more ☐ less

than the time it takes Jada. The diagram shows Andre's

travel time is ☐ 75% ☐ 175% of Jada's travel time.

- Let x represent Jada's time. Write an expression to model how long it takes Andre to get to school.
1.75x
- If it takes Jada 8 minutes to get to school, calculate how long it takes Andre to get to school.
14 minutes

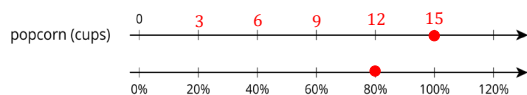
The terms "percent increase" and "percent decrease" are used to describe a specific **percent of change**.



A **percent of change** is the difference between a final value and an initial value, expressed as a percentage of the initial value.

A double number line can also be used to model percent increase or percent decrease.

- A movie theater decreased the size of popcorn bags by 20%. The old bags held 15 cups of popcorn.



- On the top number line, plot the 15 cups of popcorn as the initial value of 100%.
Shown on top number line.
- On the second line, plot 20% less than the initial 100%.
Shown on second number line.
- Use the double number line to find the number of cups in the smaller bag.
12 cups
- Discuss with your partner how the double number line is helpful for determining the impact of a 20% decrease in the size of the popcorn bag.
Answers will vary. Each segment on the number line represents 3 cups which is equivalent to 20%. A decrease of 20% shows that 3 cups need to be taken away from the initial 15 cups.

7.9.2 GUIDED INSTRUCTION: PERCENT OF CHANGE

ENRICHMENT

Consider asking students the following:

- Does it take Andre more or less time to get to school than Jada?
- What percentage is equivalent to $\frac{3}{4}$?

QUESTIONING

If time allows, consider having students share different strategies they used, or describe where they got stuck if they weren't able to determine an answer.

ENRICHMENT

If the amount of popcorn is decreased by 20%, what percentage of the original size is the new size?

7.9.3 TRY IT: PERCENT OF CHANGE

ENRICHMENT

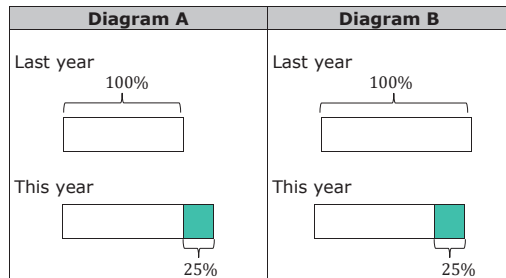
Ask students how they can distinguish between the two diagrams.

- How can you tell if the question pertains to percent increase based on the diagram?
- How can you tell if the question pertains to percent decrease based on the diagram?



7.9.3 Try It: Percent of Change

1. Explain which diagram, A or B, matches each situation.

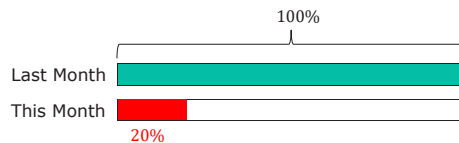


Situation	Matching Diagram	Reason
Compared to last year's strawberry harvest, this year's harvest increased 25%.	A	Diagram A matches because this year is 25% more than last year.
This year's blueberry harvest is 75% of last year's.	B	Diagram B matches because this year is 25% less than last year.
Compared to last year, this year's peach harvest decreased by 25%.	B	Diagram B matches because this year is 25% less than last year.
This year's plum harvest is 125% of last year's plum harvest.	A	Diagram A matches because this year is 25% more than last year.

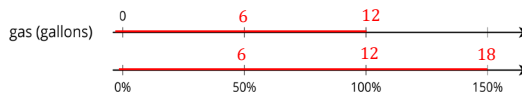
ENRICHMENT

If there were 100 mosquitoes last month, how many were there this month?

2. The number of mosquitoes decreased by 80% from last month to this month. Create a tape diagram of this month to represent the situation.



3. The gas tank in Robert's truck holds 12 gallons. The gas tank in his mom's truck holds 50% more gas. Using the double number lines, determine how many gallons the gas tank of Robert's mom's truck holds.



The gas tank in Robert's mom's truck holds 18 gallons.



7.9.4 Collaboration: Calculating Percent Increase or Decrease

Percent increase and decrease can also be found through calculations.

Complete the following with your partner.

1. A new cereal box contains “20% More” cereal. The original cereal boxes contained 500 grams of cereal.



- a. Explain how to determine how much cereal is in the new box.
Find what 20% of 500 is. Then add it to 500. This will be 120% of 500.

- b. How much cereal comes in the new box?

$$500 + 0.2(500) = 600 \text{ grams}$$

2. The price of a shirt at ClothesMart is \$18.50, but there is a store coupon that lowers the price by 20%.

- a. Work with your partner to determine how much the shirt is with the coupon.



$$18.5 - 0.2(18.5) = \$14.80$$

The table shows two different ways to complete this problem, along with corresponding answers.

Approach 1	Approach 2
The cost of the shirt with the coupon will be 80% of the original cost.	The new cost of the shirt is what's remaining after the 20% has been deducted from the original cost.
New cost = $0.8(18.50)$	New cost = $18.50 - 0.2(18.50)$

- b. Explain whether or not these equations are equivalent.
These equations are equivalent because both can be used to find the new cost correctly. In Approach 1, the discount is removed from the percentage to find 80% of the original cost. In Approach 2, the price of the discount is first calculated and then subtracted from the original cost.
- c. Did your answer to Part A match the solution(s) of the equations in the table?
Yes



$$\text{Percent of Change} = 100 \times \frac{\text{Final Value} - \text{Initial Value}}{\text{Initial Value}}$$

3. The original cost of pants at ClothesMart is \$30.00, and after a store coupon is applied, the cost is only \$25.50.

- a. Determine the percent of change by filling in the blanks of the percent of change ratio for the situation.

$$\frac{25.50 - 30}{30} = \frac{25.50 - 30}{30} = \frac{-4.5}{30} = -15\%$$

7.9.4 COLLABORATION: CALCULATING PERCENT INCREASE OR DECREASE

TEACHER MOVES

Take Turns

Ask students to take turns solving each problem and then explaining their solution. This routine ensures each and every student has the opportunity to articulate their mathematical thinking and critique the reasoning of others.

DIFFERENTIATE INSTRUCTION

There are multiple strategies for solving the problems in this component. Encourage students to use strategies that make sense for them as they build fluency in answering questions 1 and 2.

TEACHER MOVES

Algebra Talk

Ask students to compare the numeric expressions and discuss what is the same and different. How is the 20% discount represented in each?

DIFFERENTIATE INSTRUCTION

Students may struggle to understand the series of boxes in Item 3. If needed, walk through the steps with students or draw the problem on the board. Label each box with the information it should contain such as “initial cost,” “cost after coupon,” “difference,” and “percent change.” Then, have students fill the numbers in the boxes and use them to verbally explain the problem.

Be sure to provide students the opportunity to try it with their partner before jumping in to model. Consider also sharing strategies you observe on the board and asking students to compare them and determine errors.

7.9.4 COLLABORATION: CALCULATING PERCENT INCREASE OR DECREASE (CONTINUED)

- b. Explain whether the percent of change represents a percent increase or decrease based on its value.
This is a percent decrease because a discount decreases the price. You can also tell because the resulting change is a negative percentage.

- c. Complete the statements.

When calculating the percent of change using the ratio,

a positive value represents a percent ☐ increase,
☐ decrease,

and a negative value represents a percent ☐ increase.
☐ decrease.

In this context, it makes sense that the percent change is a

percent ☐ increase
☐ decrease because the change is the result of a discount.

**7.9.5 Your Turn!**

1. A school had 1,200 students last year and only 1,080 students this year. What was the percentage decrease in the number of students?

$$\frac{1080 - 1200}{1200} = \frac{-120}{1200} = -10\%$$

There was a 10% decrease in the number of students.

2. A juice box has 20% more juice in its new packaging. The original packaging held 12 fluid ounces. How much juice does the new packaging hold?

$$1.2(12) = 14.4 \text{ fluid ounces}$$

7.9.5 YOUR TURN!**QUESTIONING**

Consider connecting the context in these and other problems in the lesson to familiar or local situations to help students make connections to the problems. For example, perhaps there was a change in the number of students at their school or there is a package advertising 20% more that could be brought in for reference. Student motivation can be ignited if the problems involve people, places, and things that they are interested in.

**7.9.6 Wrap-Up: Error Analysis**

1. The number of campers who attend a local summer camp has increased from 1,938 to 2,128. The camp director calculates the percent of increase for an upcoming presentation. Her work is shown.

$$\begin{aligned} 2,128 - 1,938 &= 190 \\ 190 \div 2,128 &= 0.089 \\ 0.089 &= 8.9\% \end{aligned}$$

Explain the camp director's mistake.

The answer should be about 9.8% because $\frac{2,128-1,938}{1,938}$ is about 0.098. The camp director divided the change in number of campers by the new number instead of the original.

7.9.6 WRAP-UP: ERROR ANALYSIS**QUESTIONING**

Students who are done early can also be asked to find the correct answer.